

Document Control

- System Name: Application
- Version: 1.0
- Date: Not present in source JSON
- Author: Not present in source JSON

Executive Summary

- The Application ingests customer and order CSV data into Delta tables, performs data cleansing and computation of TotalAmount, joins data for an ordersummary table with SCD Type 2 handling, and creates a customeraggregatespend table aggregating daily spend.
- Implementation is executed using Delta Live Tables.
- All processing logic, schemas, and rules are preserved as per documented requirements.

Business Objectives

- Ingest customer and order CSV files into Delta tables for further processing.
- Cleanse data by removing "Null" or Null values and duplicate records.
- Calculate TotalAmount as a derived field in the order table.
- Join customer and order data to generate an ordersummary table, handling customer changes with SCD Type 2.
- Aggregate spend per Name and Date in a customeraggregatespend table.
- Implement the data pipeline end-to-end using Delta Live Tables.

In-Scope

- Source ingestion from defined file system locations.
- Creation and management of Delta tables: customer, order, ordersummary, customeraggregatespend.
- Dedicated data cleansing actions.
- Calculation of TotalAmount.
- Join logic and SCD Type 2 support for ordersummary.
- Aggregation from ordersummary to customeraggregatespend.
- Implementation through Delta Live Tables.

Out-of-Scope

- Integration of data sources not specified in the requirements.
- Security controls, SLAs, monitoring, and alerting features.
- Performance tuning, scaling, and optimization beyond stated actions.
- Scheduling, orchestration, and DevOps mechanisms except execution via Delta Live Tables.

- Analytics or reporting interfaces.

Stakeholders

- Not present in source JSON

Business Requirements

BR-APP-001 — Source Data Ingestion

- Description: Load customer and order CSV files from defined filesystem paths into Delta format tables.
- Business Value: Enables a structured, governed raw data layer.
- Priority: High

BR-APP-002 — Table Schemas Standardization

- Description: Enforce specific schemas for customer (CustId, Name, EmailId, Region) and order (OrderId, ItemName, PricePerUnit, Qty, Date, CustId) tables.
- Business Value: Ensures consistent structure for all downstream operations.
- Priority: High

BR-APP-003 — Computed Field: TotalAmount

- Description: Add TotalAmount field to order table calculated as $\text{PricePerUnit} \times \text{Qty}$.
- Business Value: Supports accurate spend measurements and reporting.
- Priority: High

BR-APP-004 — Data Cleansing

- Description: Remove "Null" and Null literals and deduplicate records in both customer and order tables.
- Business Value: Increases data reliability and trustworthiness for analytics.
- Priority: High

BR-APP-005 — Ordersummary Table Creation

- Description: Join customer and order tables on CustId to form an ordersummary table, applying SCD Type 2 change tracking.
- Business Value: Maintains complete audit trail of customer data changes in transactions.
- Priority: High

BR-APP-006 — SCD Type 2 Handling Logic

- Description: Process changes in customer attributes using SCD Type 2 approach, updating previous records to Inactive and new records to Active, with StartDate and EndDate tracking.

- Business Value: Retains historical versions, critical for compliance and audit.
- Priority: High

BR-APP-007 — Aggregate Spend Calculation

- Description: Aggregate ordersummary data to calculate daily spend per customer in the customeraggregatespend table using sum(TotalAmount), grouped by Name and Date.
- Business Value: Enables business trend analysis and spend monitoring.
- Priority: Medium

BR-APP-008 — Pipeline Implementation with Delta Live Tables

- Description: All processing steps (1–7) are implemented using Delta Live Tables pipeline.
- Business Value: Ensures automation, reliability, and scalability for production use.
- Priority: High

Assumptions & Constraints

- Only requirements explicitly present in the source JSON are assumed.
- Delta Live Tables is the designated implementation technology.
- SCD Type 2 attribute field names (StartDate, EndDate, Active) are referenced for logic but not present in explicit field lists.

Success Metrics

- 100% of source records are processed with removal of all "Null"/Null and duplicates.
- Accurate computation of TotalAmount.
- All customer changes logged with SCD Type 2 in ordersummary.
- customeraggregatespend generated with correct daily aggregation across all relevant customers.
- End-to-end process successfully orchestrated using Delta Live Tables.

Risks & Mitigations

- Ambiguity on explicit definitions for StartDate, EndDate, and Active fields in SCD Type 2: Clarification required for precise schema mapping.
- Data quality issues could remain if additional checks outside of "Null" and duplicate removal are needed: Future enhancement may be needed based on actual data variance.
- Lack of detail on error handling, retry, and monitoring: Suggests need for incremental improvement post-initial build.

Approval

- Not present in source JSON

Appendix

Appendix A: Source Definitions

Source Name	Type	Location	Format	Notes
Customer source	CSV	/Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/customerdata	CSV	Loaded into Delta table customer
Order source	CSV	/Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/orderdata	CSV	Loaded into Delta table order

Appendix B: Target Definitions

Layer	Table	Columns	Notes
Raw	customer	CustId, Name, EmailId, Region	Loaded from Customer
Raw	order	OrderId, ItemName, PricePerUnit, Qty, Date, CustId	Loaded from Order s
Silver	ordersummary	CustId, Name, EmailId, Region, OrderId, ItemName, PricePerUnit, Qty, Date	Join of customer and
Gold	customeraggregatespend	Name, TotalAmount, Date	Aggregate spend by

Appendix C: Transformations

Step	Transformation	Input	Output	Logic
1	Add computed column	order	order.TotalAmount	PricePerUnit × Qty
2	Remove "Null"/Null	customer/order	customer/order	Remove all records with null-like values
3	Deduplicate records	customer/order	customer/order	Drop exact duplicate records
4	Join tables	customer, order	ordersummary	Join on CustId, SCD Type 2 handling
5	Aggregate spend	ordersummary	customeraggregatespend	Group by Name, Date; sum TotalAmount

Appendix D: Business Rules

Rule ID	Area	Description	Source
BR-1	Computation	TotalAmount = PricePerUnit × Qty	Requirements
BR-2	Cleansing	Remove "Null" and duplicate records	Requirements
BR-3	Join	Join customer to order on CustId	Requirements
BR-4	SCD Type 2	Maintain inactive/active and StartDate/EndDate	Requirements
BR-5	Aggregation	Group and sum to customeraggregatespend	Requirements

Appendix E: Process Flow

Step	Description	Dependency
1	Load customer/order CSV to Delta table	Source availability
2	Cleanse and deduplicate data	Raw tables created
3	Add computed column TotalAmount to order	order table exists
4	Join customer and order to create ordersummary with SCD Type 2	Both tables cleansed and ready

5	Group ordersummary to generate customeraggregatespend	ordersummary available
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Appendix F: Entity Relationship Diagram (ERD)

[customer]

*CustId

Name

EmailId

Region

[order]

*OrderId

ItemName

PricePerUnit

Qty

Date

+CustId

TotalAmount

[ordersummary]

*CustId

Name

EmailId

Region

OrderId

ItemName

PricePerUnit

Qty

Date

[customeraggregatespend]

*Name

TotalAmount

Date

customer 1--* order

customer *--* ordersummary

ordersummary *--* customeraggregatespend